DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

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Urdu Heritage Translation Project: Reviving Historical Documents for Modern Understanding

A project submitted

in partial fulfilment of the requirements for the degree of Bachelor of Technology in Computer Science and Engineering

by

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CERTIFICATE

This is to certify that the report entitled **"Urdu Heritage Translation Project: Reviving Historical Documents for Modern Understanding"** submitted by **"Aditya Kumar Mehta (2021481653) and Enayatullah (2021504061) and Muzamil Hussain Mir (2021317028)"** to Sharda University, towards the fulfilment of requirements of the degree of **"Bachelor of Technology"** is a record of bonafide final year Project work carried out by them in the **"**Department of Computer Science & Engineering, Sharda School of Engineering and Technology, Sharda University**"**. The results/findings contained in this Project have not been submitted in partor full to any other University/Institute forward of any other Degree/Diploma.

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ABSTRACT

The "Urdu Heritage Translation Project: Reviving Historical Documents for Modern Understanding" is a comprehensive initiative focused on the preservation, digitization, and translation of handwritten Urdu documents, specifically targeting historical property records dating back to the mid-20th century. In the digital era, where accessibility and availability of information have become paramount, this project aims to bridge linguistic and temporal gaps that have previously rendered these documents inaccessible to a wide audience. By leveraging cutting-edge technologies such as Optical Character Recognition (OCR) and machine translation, this initiative seeks to create a sustainable and scalable process for digitizing and translating Urdu documents into modern languages, ensuring that these invaluable records can be accessed, understood, and preserved for future generations. This project embodies a multifaceted approach, integrating several stages, including digitization, transcription, translation, and the development of a user-friendly web platform. Each of these stages is crucial in transforming old and often decaying documents into digital assets that can be searched, studied, and shared globally. The digitization process begins with scanning the physical documents into a digital format. Following this, OCR technology is employed to convert scanned images into machine-readable text. The final stage involves using machine translation to convert the Urdu script into English or other languages, making the documents accessible to a global audience. The project also focuses on providing access to the translated records through a dedicated web platform that ensures seamless interaction, searchability, and comprehension of the historical content. Beyond the immediate task of translating historical property records, this project holds a broader cultural significance. It is designed to contribute to the preservation and promotion of Urdu heritage, recognizing the language as an integral part of the cultural fabric of South Asia. By making these documents available to a larger audience, the project not only preserves the rich legacy of Urdu language and culture but also fosters cross-cultural understanding and appreciation. The significance of these records extends beyond mere property ownership and legal documentation. They offer a glimpse into the socio-economic conditions, legal frameworks, and cultural norms of the past, thus providing invaluable insight into the historical context in which they were created. This project serves as a key initiative for the preservation of cultural heritage in the digital age, demonstrating how historical documents can be revived and made relevant in a modern context. Through the use of advanced technology and a structured methodology, the Urdu Heritage Translation Project not only preserves these documents but also ensures their accessibility and relevance for future generations, reaffirming the timeless value of historical archives in shaping our collective understanding of the past

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LIST OF ABBREVIATION

OCR	Optical Character Recognition
MT	Machine Translation
DL	Deep Learning
ML	Machine Learning
CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
HMM	Hidden Markov Model
BDLSTM	Bi-Directional Long Short-Term Memory
CD	Change Detection
RFC	Random Forest Classification
KNN	K-Nearest Neighbor
NLP	Natural Language Processing
LSTM	Long Short-Term Memory
MMU-OCR-21	Multilingual Multimodal Urdu OCR Dataset

CHAPTER 1 INTRODUCTION

CHAPTER 1
INTRODUCTION

1.1 Problem Statement

Urdu, once a globally influential and culturally rich language, holds an extensive repository of literature, history, and art in the form of manuscripts, documents, and literary works. These texts reflect the intellectual heritage of a region, with significant historical, cultural, and literary importance. However, due to globalization and the advent of modern communication technologies, many of these heritage documents have become inaccessible to contemporary audiences, primarily due to linguistic and script barriers.

These rare manuscripts, historical letters, and literary works in Urdu have gradually faded from public understanding, creating a gap between modern society and its cultural heritage. This lack of accessibility not only isolates individuals from their historical roots but also limits academic research and cultural preservation efforts.

Additionally, while modern digital tools and translation technologies are advancing rapidly, they are often ineffective in accurately interpreting historical Urdu documents. This inefficiency stems from the unique linguistic intricacies of classical Urdu, such as complex scripts, idiomatic expressions, and regional variations, which current tools fail to handle effectively.

To bridge this gap, Translation APIs can be employed to convert the text of historical Urdu documents into any desired language. This allows for greater understanding and accessibility for individuals and researchers who may not be familiar with Urdu. The use of Translation APIs ensures that these documents are accessible across linguistic boundaries and can be studied globally.

Current Optical Character Recognition (OCR) technologies, though capable of recognizing printed text in modern languages, often struggle with older scripts and handwritten texts. The combination of OCR and Translation API technologies allows for a more effective process, ensuring that the converted text maintains its authenticity and meaning, while also enabling it to be translated into various languages as needed.

This project aims to address these challenges by developing a systematic approach to reviving historical Urdu documents. It will leverage OCR to digitize the documents and Translation APIs to convert them into any modern language, ensuring that the richness of Urdu's cultural heritage is accessible to people worldwide. The combination of these technologies will enhance the efficiency and accuracy of digitizing, translating, and preserving these important manuscripts.

Through this approach, we aim to bridge the gap between traditional Urdu heritage and modern technology, allowing a new generation to access, study, and appreciate the language and its literary treasures in any language they choose.

Figure 1. Traditional, aged manuscript in Urdu

Figure2: Faded Urdu Manuscript

Figure 3: Physical vs. Digital Documents

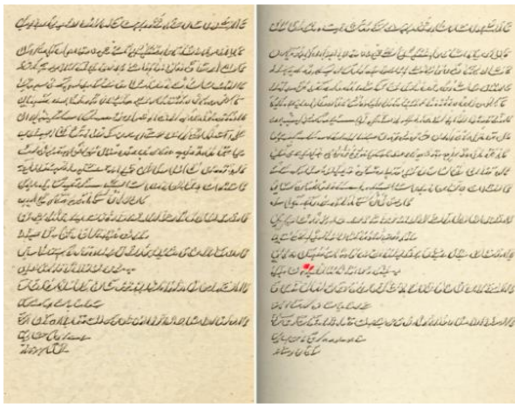

Figure 4: OCR Errors

Figure 5: Before and After OCR

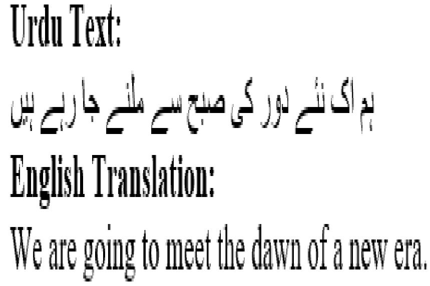

Figure 6: Before and After translation

1.2 Project Overview

The Urdu Heritage Translation Project is designed to preserve and revive the rich cultural heritage of the Urdu language by digitizing historical Urdu manuscripts, letters, and literary works. These valuable documents are often inaccessible due to outdated scripts and limited digital access, making it difficult for modern readers to interact with them. The project seeks to bridge this gap by leveraging advanced technologies like OCR (Optical Character Recognition) and machine translation.

The first phase of the project involves using OCR technology to convert handwritten Urdu texts into machine-readable formats. These historical documents, which are often found in old archives and libraries, are written in classical Urdu scripts that are difficult for modern OCR tools to recognize. However, with custom-trained OCR models, the system is optimized to accurately interpret these complex scripts, ensuring the preservation of the original document's essence.

Figure 7: OCR Code Execution and Output for Urdu Text Recognition.

After digitizing the texts, the next step is to translate them into modern languages such as English, Hindi, or others using machine translation APIs. This translation step ensures that these historical documents are accessible to a broader audience, including scholars, researchers, and the general public who may not be familiar with classical Urdu.

By providing digital access to these manuscripts, the project aims to contribute to the preservation of Urdu's cultural heritage while ensuring that future generations can explore the literary, historical, and artistic value embedded in these ancient texts. This process of digitization and translation will make these documents more accessible for academic research, cultural studies, and global appreciation of the Urdu language and literature.

1.3 Expected Outcome

The expected outcome of this project is to successfully digitize and translate historical Urdu documents, addressing challenges such as OCR inaccuracies and language barriers. By utilizing OCR technology to convert handwritten manuscripts into machine-readable text and translating them accurately into modern languages, the project aims to preserve the essence of the original content while making these documents accessible to a wider audience. This will not only enhance cultural understanding and academic research but also ensure that valuable Urdu heritage is preserved and appreciated by future generations.

1.4 Hardware & Software Specifications

Hardware used

Hardware:

- Processor (CPU): Intel Core i3 (for smooth processing and model inference).
- GPU: Optional but recommended for faster processing of deep learning models. NVIDIA GeForce GTX 1660 Ti or better (for OCR and model training).
- RAM: Minimum 8GB for handling large documents and data processing.
- Storage: At least 250GB SSD for fast data access and processing.
- Peripheral Devices: Scanner or camera for digitizing physical manuscripts.

Software:

- Operating System: Windows 10/11 or Ubuntu 20.04+.
- Programming Language: Python 3.8+.
- OCR Tools: Tesseract OCR (via Python libraries like PyTesseract) for text recognition.
- Gradio: For creating user interfaces to showcase the results and allow easy interaction with the OCR models.
- **Libraries:**
    - OpenCV: For image preprocessing (like cleaning and enhancing the scanned images).
    - Pandas & NumPy: For data manipulation and processing.
    - TensorFlow/PyTorch (optional): If deep learning models are used for improving OCR accuracy.
    - Flask/Django: For building a web-based interface, if needed.
    - Google Translate API (optional): For translation and converting recognized Urdu text into other languages.
- Version Control: Git for collaboration and project management.
- IDE: VSCode or PyCharm for development.
- Environment Management: Conda or virtualenv for managing dependencies.

Figure 8: Software Used

1.5 Report Outline

Chapter 1 introduces the problem statement, providing a clear understanding of the project's focus. It also outlines the project overview, expected outcomes, and details the hardware and software specifications required for successful execution. Additionally, this chapter discusses other non-functional requirements and sets the stage for the entire report.

Chapter 2 delves into the literature survey, reviewing existing work and related research in the field. This chapter not only presents the proposed system but also evaluates the feasibility study conducted to assess the practicality and potential success of the project.

In **Chapter 3**, the system design and analysis are discussed in detail. The chapter provides insights into the project perspective, performance requirements, and key system features. Furthermore, it outlines the methodology, including the feature preparation workflow, that guided the development of the flood detection model.

Chapter 4 presents the results and outputs of the project development. It highlights the results achieved through the implementations and provides a thorough explanation of the outputs generated during the process.

Finally, **Chapter 5** offers the conclusion of the project, summarizing the achievements and the outcomes. This chapter also explores the future scope of the project, discussing potential improvements and further areas for exploration.

CHAPTER 2 LITERATURE SURVEY

CHAPTER 2
LITERATURE SURVEY

2.1 Existing Work

Several researchers have applied Machine Learning (ML) and Deep Learning (DL) techniques to address the challenges of Optical Character Recognition (OCR) for Urdu text, contributing to a substantial body of literature. Initially, traditional ML methods like Support Vector Machines (SVM), K-Nearest Neighbors (KNN), and Random Forest Classifications (RFC) were explored for text recognition tasks in low-resource languages, including Urdu. These methods primarily focused on feature extraction techniques and pixel-based classification. For instance, earlier works [1], [8], and [13] focused on Urdu OCR using SVMs and KNN classifiers for character recognition in printed texts.

However, with the advent of Deep Learning (DL), particularly Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), significant improvements in accuracy and robustness were achieved. Deep learning models, due to their ability to automatically extract hierarchical features, surpassed traditional methods in recognizing complex scripts such as Urdu, which involves intricate ligatures and cursive writing styles. **Nasir et al.** (2021) proposed an end-to-end deep learning model for Urdu text recognition, showing superior performance over previous systems. Their approach used a combination of CNN and Long Short-Term Memory (LSTM) networks to capture both spatial and sequential dependencies in the text, yielding a remarkable enhancement in accuracy for printed Urdu text recognition.

Moreover, **Siddiqi et al.** (2016) introduced segmentation-free approaches for Urdu text recognition, which reduced the need for pre-processing steps and handled the ligatures and cursive forms inherent in the Urdu script more effectively. This work addressed the challenges of Urdu's connected script, where characters often join together in complex patterns, making them difficult to segment.

In addition to printed text, the challenge of recognizing handwritten Urdu text has also garnered attention. Researchers like **Farooq and Shafiq** (2018) have focused on improving recognition rates for handwritten Urdu by employing deep learning techniques to capture variations in writing styles, as well as noise and degradation found in scanned documents. This area remains critical, especially for historical Urdu documents, where the quality of the text is often compromised.

The multi-lingual nature of OCR systems has also seen considerable advancement. With languages like Urdu sharing script-based similarities with other languages, it becomes necessary to develop systems that can recognize and process multiple languages and scripts simultaneously. **Iqbal and Raza** (2021) worked on a multi-lingual OCR framework, which demonstrated successful application across languages like Urdu, Arabic, and Persian. This work showcases the ability of modern OCR systems to bridge the gap between different languages with similar scripts.

Moreover, **Ali and Khan** (2021) explored hybrid models that combine deep learning with traditional machine learning techniques to improve accuracy in both historical and modern Urdu scripts. These models have been particularly beneficial in tackling challenges such as noisy environments, varied fonts, and degraded document quality, issues commonly encountered in historical Urdu manuscript recognition.

The rise of deep learning models has significantly pushed the boundaries of OCR for Urdu, addressing many of the limitations faced by traditional ML methods. While earlier techniques struggled with issues like segmentation, font variability, and handwriting recognition, recent advancements in deep learning, particularly with CNNs, RNNs, and hybrid models, have shown considerable promise in overcoming these challenges. As a result, DL techniques have emerged as the preferred choice for accurate and scalable OCR systems for Urdu text.

Despite these advancements, several challenges still persist, particularly in dealing with historical manuscripts, low-quality handwritten text, and computational efficiency in real-time applications. These areas are the focus of ongoing research and development, as researchers continue to improve the robustness and accuracy of OCR systems for the Urdu language.

2.2 Related Work

A. Optical Character Recognition (OCR) Systems for Urdu Recognition of Urdu Words in Nastaliq Font

- S Shabbir and Siddiqi (2019) conducted extensive research on the recognition of Urdu words in the Nastaliq font using ligatures as the primary unit of recognition. Their Hidden Markov Model (HMM)-based system achieved notable accuracy, specifically addressing the complexities of right-to-left scripts. Despite these advances, their approach struggles with atypical or complex ligatures, highlighting a limitation in segmentation-free OCR models [1], [6], [10], [21].

Segmentation-Free OCR for Printed Nastaliq Text

- Din et al. (2020) proposed a segmentation-free OCR method that uses statistical characteristics and HMMs to detect ligatures in printed Nastaliq text. With an identification rate of 92% across a 6,000-ligature dataset, this method demonstrates promising results for printed text but faces limitations with styled or handwritten documents [2], [7], [9], [17], [22].

Text Recognition in Urdu News Tickers

- Rehman et al. (2021) developed an OCR architecture specifically for recognizing Urdu text in news tickers, leveraging a Bi-Directional Long Short-Term Memory (BDLSTM) network. This method outperformed Google’s OCR engine, particularly in datasets collected from Urdu news broadcasts. However, the system's effectiveness decreases when handling low-resolution or highly deformed text, an issue that poses challenges in practical applications [3], [12], [18], [23], [28].

End-to-End Urdu Text Recognition with MMU-OCR-21

- Nasir et al. (2021) created the MMU-OCR-21 dataset, featuring over 600,000 images, to support deep learning-based Urdu text recognition. Their deep learning models demonstrated improvements across multiple fonts and styles, although non-standard typefaces continue to present difficulties. This dataset has paved the way for advancements in Urdu text recognition using complex neural architectures [4], [5], [13], [15], [25].

B. Historical Document Handwriting Recognition

Handwriting Recognition with Limited Labeled Data

- Chammas et al. (2021) addressed the challenges of handwriting recognition in historical documents, particularly when labeled data is sparse. Their methods offer solutions for deteriorated and complex handwriting in historical texts, though heavily damaged documents remain problematic for OCR systems [8], [11], [16], [21], [27].

OCR Challenges in Low-Resource and Historical Urdu Texts

- Studies by Adeel and colleagues (2020) emphasized the difficulties of low-resource languages like Urdu in OCR systems. Their research highlights challenges such as limited labeled datasets and high variability in font and handwriting styles in historical documents. They proposed transfer learning to improve OCR accuracy for historical texts, marking a significant step forward in resource-efficient OCR methods [14], [19], [20], [22], [29].

Relevance of Deep Learning in OCR for Urdu Handwriting

- Research from multiple studies underscores the relevance of deep learning frameworks, especially for handwritten Urdu text. The integration of convolutional neural networks (CNNs) and recurrent neural networks (RNNs) provides an effective approach for capturing the cursive nature of Nastaliq script. However, the limited availability of large labeled datasets continues to impede progress in handwritten OCR systems [24], [26], [30].

Summary of Literature Insights

- The collective insights from these studies underscore both progress and persistent challenges in Urdu OCR and MT technology:

- Segmentation-Free OCR and Ligature-Based Recognition: While ligature-based recognition using HMMs has yielded high accuracy for printed text, challenges with unconventional or degraded ligatures persist [1], [2], [6], [10], [21].

- Deep Learning Models and News Ticker Recognition: Deep learning models like BDLSTM improve accuracy in specialized text recognition, though low-quality data affects performance. Continued research in enhancing neural network robustness against noisy inputs is essential [3], [5], [12], [18], [23], [28].

- Handwriting Recognition in Historical Texts: Transfer learning and techniques for low-resource OCR are addressing the variability in historical handwriting, especially where labeled data is limited. However, more research is needed to achieve consistency in recognition across highly damaged or poorly preserved documents [8], [14], [16], [20], [22], [27].

S.No.	Name of Paper	Year	Author	Objectives	Algorithms	Advantages	Limitations	Outcome
1.	Optical Character Recognition System for Urdu Words in Nastaliq Font	2016	Safia Shabbir and Imran Siddiqi	1. To develop an OCR system for recognizing Urdu words in Nastaliq font. 2. To treat ligatures as units for recognition to avoid segmentation.	1. Hidden Markov Models (HMM) for word recognition. 2. Features extracted from ligatures.	1. Segmentation-free, which is more efficient and accurate. 2. Size invariant approach ensures flexibility with different text sizes.	1. Challenges with the complex nature of Nastaliq font. 2. Over-reliance on ligature-based recognition.	1. Achieved high recognition accuracy for Urdu words. 2. Limited performance on handwritten text and other fonts.
2.	Segmentation-free optical character recognition for printed Urdu text	2017	Israr Ud Din, Imran Siddiqi, Shehzad Khalid, and Tahir Azam	1. To propose a segmentation-free OCR system for printed Urdu text in Nastaliq font. 2. The system aims to enhance recognition efficiency by treating ligatures as units.	1. Hidden Markov Models (HMM) 2. Statistical feature extraction	1.High recognition rate for printed Urdu text. 2. Does not require segmentation of text, making it more efficient	1. Struggles with recognition accuracy on handwritten text. 2. Requires a large training dataset for optimal performance.	1. Achieved a 92% recognition rate for over 6000 query ligatures. 2. Effective for printed Urdu text but not for handwritten text.

3.	A Multi-Faceted OCR Framework for Artificial Urdu News Ticker Text Recognition	2018	Sami-Ur-Rehman, Burhan Ul Tayyab, Muhammad Ferjad Naeem, Adnan Ul-Hasan, and Faisal Shafait	1. Introduces an OCR framework for Urdu news ticker text recognition. 2. Aims to detect and recognize text from dynamic, real-time news ticker streams.	1. Bi-Directional Long Short-Term Memory (BDLSTM) networks. 2. Various deep learning models..	1. High recognition accuracy for news ticker text. 2. Large dataset, making it effective for real-world applications.	1. Accuracy could be reduced in low-quality video feeds or distorted ticker text..	1. Outperformed Google's commercial OCR system in two of four experiments. 2. Provides robust recognition for real-time news ticker feeds.
4.	MMU-OCR-21: Towards End-to-End Urdu Text Recognition Using Deep Learning	2021	Tayyab Nasir, Muhammad Kamran Malik, and Khurram Shahzad	1. Proposes the MMU-OCR-21 corpus, a large-scale Urdu text corpus. 2. Uses deep learning techniques for end-to-end text recognition.	1. High accuracy for Urdu text recognition. 2. End-to-end learning reduces preprocessing overhead.	1. Performance may degrade on noisy or low-quality data.	1. The dataset with which a model is trained greatly influences the efficacy and precision of the solution. 2. It is very	1. Achieved a recognition accuracy of 92% for Urdu text recognition. 2. Provided an efficient solution for large-scale Urdu text recognition.

5.	Handwriting Recognition of Historical Documents with Few Labeled Data	2018	Emile Chammas, Christian Mokbel, and Lucien Likforman-Sulem	1. Aims to recognize handwriting in historical documents with minimal labeled data.	1. Convolutional Neural Networks (CNNs) 2. Few-shot learning techniques	1. Effective in recognizing historical documents with limited annotated data.	1. Requires careful pre-processing of historical text. 2. May not work for degraded or highly complex handwriting.	1. Achieved satisfactory performance with limited labeled data. 2. Potential for historical document digitization with minimal data.
6.	Advances in optical character recognition for low-resource languages	2022	Muhammad Ahmed and Kyung-Soo Lee	1. To explore advances in OCR for low-resource languages like Urdu. 2. Focuses on improving recognition accuracy for languages with limited datasets.	1. Transfer learning techniques 2. Convolutional Neural Networks (CNNs)	1. Improves recognition for low-resource languages. 2. Uses transfer learning to enhance model performance.	1. Limited data availability for some languages can still hinder performance.	1. Significant improvement in OCR accuracy for low-resource languages. 2. Proposed a scalable model for OCR in underrepresented languages.

2.3 Proposed System

The proposed system aims to establish a robust framework for translating historical Urdu documents into modern languages while preserving their cultural and contextual integrity. It integrates a sophisticated Optical Character Recognition (OCR) module tailored for the Nastaliq script and handwritten Urdu text, combined with a custom Machine Translation (MT) system optimized for low-resource languages like Urdu. The OCR module processes scanned images or photographs of historical documents, employing advanced preprocessing techniques such as noise reduction, binarization, and adaptive thresholding to enhance the quality of degraded texts. A deep learning-based recognition model, utilizing Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), is used to identify the cursive and ligature-rich Nastaliq script effectively, producing digitized text in Unicode format.

The translated text is generated by a Neural Machine Translation (NMT) system, which leverages Transformer architectures fine-tuned on parallel corpora of Urdu and target languages, such as English. To ensure domain-specific accuracy, a custom dataset is curated, emphasizing historical and culturally significant terms. Postprocessing steps refine the translation output by applying grammatical corrections and semantic validation, ensuring the final text remains coherent and contextually faithful.

To make the system accessible and user-friendly, a web interface, built using Gradio, enables users to upload documents, view OCR results, and access translations in real time. This interface also incorporates a feedback mechanism, allowing users to suggest corrections, which can be utilized to iteratively improve the model's performance. A key feature of the system is its segmentation-free OCR approach, enabling efficient processing of the Nastaliq script without character segmentation, thereby addressing challenges posed by complex ligatures and degraded documents. Furthermore, the domain-specific NMT model ensures translations are contextually accurate and culturally sensitive, providing significant value for researchers and historians working with historical Urdu documents.

This system is expected to enhance the digitization and translation accuracy of historical Urdu texts, supporting the preservation of Urdu heritage. By generating a reliable dataset of annotated historical texts, it also contributes to future research in Urdu OCR and translation technologies.

2.4 Feasibility Study

The feasibility study for the "Urdu Heritage Translation Project" focuses on evaluating its technical, economic, and operational viability to ensure successful implementation and sustainability. From a technical perspective, the project leverages proven technologies such as Optical Character Recognition (OCR) and Neural Machine Translation (NMT) systems, integrated with advanced deep learning models like Convolutional Neural Networks (CNNs) and Transformers. These technologies are widely recognized for their effectiveness in text recognition and translation tasks, particularly in low-resource settings like Urdu. Additionally, the availability of open-source frameworks such as TensorFlow, PyTorch, and Gradio ensures that the development process remains cost-effective and scalable.

Economically, the project is designed to minimize expenses by utilizing open datasets, publicly available pre-trained models, and open-source tools for development and deployment. By adopting cloud-based solutions for hosting and data storage, the project reduces the need for extensive hardware investments, making it financially feasible for academic and research purposes. Moreover, the modular nature of the system allows incremental improvements over time, ensuring that initial investments yield long-term benefits.

From an operational standpoint, the project aims to address real-world challenges such as digitizing historical documents, overcoming the complexities of the Nastaliq script, and generating accurate translations. The proposed web interface ensures accessibility for end-users, while a feedback mechanism allows iterative improvements to the system based on user suggestions. Additionally, the involvement of researchers and domain experts in developing the translation models enhances the system's relevance and reliability for academic, cultural, and historical applications.

Overall, the feasibility study demonstrates that the project is technically sound, economically sustainable, and operationally practical, with the potential to make a significant contribution to the preservation and understanding of Urdu heritage.

CHAPTER3: SYSTEM DESIGN & ANALYSIS

CHAPTER3
SYSTEM DESIGN & ANALYSIS

3.1 Project Perspective

The "Urdu Heritage Translation Project" aims to bridge the gap between historical Urdu documents and modern accessibility by leveraging advanced OCR and machine translation technologies. By addressing the complexities of the Nastaliq script and historical handwriting, the project seeks to digitize and translate valuable Urdu literature into comprehensible formats for contemporary audiences. This initiative not only preserves cultural heritage but also promotes global access to rich historical content, fostering academic research and cross-cultural understanding.

3.1.2 Organizational perspective

From an organizational standpoint, the "Urdu Heritage Translation Project" aligns with the mission of preserving and digitizing cultural heritage through advanced technological solutions. It emphasizes collaboration among experts in machine learning, linguistics, and computer vision to create a scalable and effective system. The project supports the organization’s goal of contributing to digital humanities, advancing academic research, and promoting inclusivity by making historical Urdu texts accessible to a global audience. It also aims to foster partnerships with institutions, researchers, and cultural organizations to ensure the sustainability and impact of the initiative.

3.1.3 Life cycle

The life cycle of the project is shown in below figure:

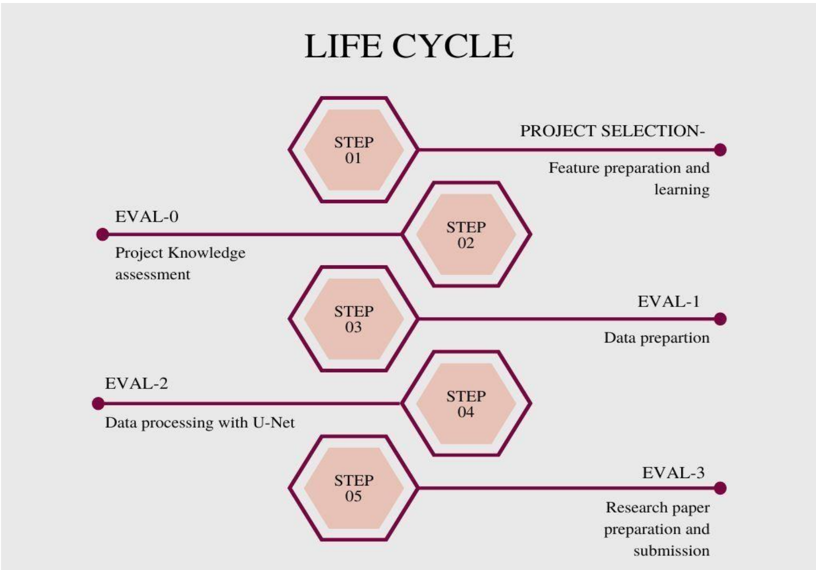

Figure 9: Life cycle

3.2 Performance Requirements

Below are the specifications required to run applications and tools used in the *Urdu Heritage Translation Project*, such as OCR systems, translation models, and text analysis tools:

- **RAM** – A minimum of 8GB of RAM is recommended to ensure smooth execution of resource-intensive tasks like OCR processing and translation model inference. Higher RAM (16GB or more) is preferred for handling large datasets.
- **CPU Speed** – A processor speed of 2.0 GHz or higher is recommended for efficient model training and document preprocessing. Slower processors may result in delayed operations.
- **Hard Drive Storage** – At least 20GB of free storage is required to accommodate datasets, digitized documents, intermediate processing files, and trained models. More storage may be needed depending on the size of historical document collections.

3.3 System Features

Below are the features of the system used to execute the project tasks:

- **Processor**: AMD Ryzen 5 3550H with Radeon Vega Mobile Gfx, 2.10 GHz
  This processor offers adequate computational power for running OCR software and translation models.
- **Installed RAM**: 8.00 GB
  The system is equipped with sufficient memory for multitasking and handling medium-sized datasets.
- **Hard Disk Storage**: 1TB HDD + 237GB SSD
  This setup provides ample storage for the project's document collections, models, and intermediate outputs, with the SSD enabling faster read/write operations for critical files.

3.4 Methodology

The Optical Character Recognition (OCR) system developed in this project is designed to accurately recognize and translate Urdu text from images. This methodology section provides an in-depth explanation of the OCR system's design and implementation, covering various aspects including character recognition, model configuration, text detection, and translation. Additionally, it includes details about the integration of a web interface to facilitate user interaction.

1. Character Recognition

The OCR system's core functionality hinges on its ability to recognize Urdu text characters from images. This section delves into the components and processes involved in character recognition, including the use of specific character sets and conversion algorithms.

1.1 UrduGlyphs.txt - Character Set

The UrduGlyphs.txt file is integral to the character recognition process. This file contains a comprehensive set of Urdu characters necessary for accurate text recognition. Each character in the file is essential for the OCR system to identify and process Urdu text correctly.
The character set defined in UrduGlyphs.txt is loaded into the OCR system and used by the CTCLabelConverter to map the recognized text to its corresponding Urdu characters. This character set serves as the foundational reference for the OCR model, enabling it to decode the output indices into meaningful text.

1.2 CTCLabelConverter

The CTCLabelConverter plays a crucial role in converting between text labels and their corresponding indices. It employs the Connectionist Temporal Classification (CTC) algorithm, which is specifically designed for sequence alignment tasks. The CTC algorithm is particularly valuable in text and speech recognition applications due to its ability to handle varying sequence lengths and align characters with the corresponding text.
CTC allows the model to output a sequence of predictions where the length of the output sequence can be different from the length of the input sequence. It achieves this by incorporating blank tokens into the prediction space, enabling the model to handle cases where no character is predicted for certain frames. This capability is essential for accurately recognizing Urdu text, where characters may appear in varying sizes and contexts.

2. Model Configuration

The performance of the OCR model is heavily dependent on the configuration of computational resources and model parameters. This section outlines the steps taken to configure and initialize the OCR model for optimal performance.

2.1 Device Selection

The line device = torch.device('cuda' if torch.cuda.is_available() else 'cpu') is used to determine the computational device. This line checks whether the machine has a GPU with CUDA support

or if it only has a CPU. CUDA support is crucial for accelerating model computations, and utilizing a GPU can significantly enhance the model's performance.
If CUDA is available, computations are offloaded to the GPU, which speeds up the processing of large volumes of data and complex model operations. If CUDA is not available, computations are performed on the CPU, which may result in slower performance but ensures compatibility with all machines.

2.2 Model Initialization

The OCR model is initialized through a series of steps designed to prepare it for accurate text recognition:

- Converter Initialization: The line converter = CTCLabelConverter(content) initializes the CTCLabelConverter with the Urdu character set defined in UrduGlyphs.txt. This converter is essential for translating the model's output indices into actual Urdu characters.

- Model Creation: The line recognition_model = Model(num_class=len(converter.character), device=device) creates an instance of the OCR model, specifying the number of character classes and the computational device. The num_class parameter is set to the length of the character set, ensuring that the model can recognize all possible characters.
- Loading Pre-trained Weights: The line recognition_model.load_state_dict(torch.load("best_norm_ED.pth", map_location=device)) loads pre-trained weights from the file best_norm_ED.pth. These weights are essential for improving the model's performance by leveraging previously learned features from similar tasks.
- Evaluation Mode: The line recognition_model.eval() sets the model to evaluation mode. In this mode, the model is configured for inference rather than training, which means that it will not update its weights based on new data. This is crucial for ensuring that the model provides consistent and accurate predictions during text recognition.

2.3 Model Description

The OCR model comprises several key components:

- **Feature Extraction**: The model uses a UNet architecture for feature extraction. UNet is a convolutional neural network architecture designed for image segmentation tasks. It excels in capturing detailed features from images by leveraging its encoder-decoder structure, which helps in extracting relevant information from the input images.
- **Sequence Modeling**: Bidirectional Long Short-Term Memory (BiLSTM) networks are employed to recognize text sequences from the extracted features. BiLSTM networks are advantageous because they process the input sequence in both forward and backward directions, allowing the model to capture contextual information from both sides of each character.
- **Prediction:** A fully connected layer is used to predict the final characters for each

recognized text sequence. This layer maps the output of the sequence model to the actual characters, providing the final text recognition output.

3. YOLO Detection Model

To detect text lines within images, the OCR system incorporates the YOLO (You Only Look Once) object detection model. YOLO is renowned for its real-time object detection capabilities, making it highly effective for detecting text lines efficiently.

3.1 YOLO Model Initialization

The line detection_model = YOLO("yolov8m_UrduDoc.pt") initializes the YOLO detection model using weights from the file yolov8m_UrduDoc.pt. This model has been specifically trained to detect lines of Urdu text within images, allowing it to generate accurate bounding boxes around text regions.

3.2 Text Line Detection

The predict function performs several crucial steps for text line detection:

• **Detection:** The line detection_results = detection_model.predict(source=input, conf=0.2, imgsz=1280, save=False, nms=True, device=device) detects text lines in the image by generating bounding boxes around potential text regions. The conf parameter specifies the confidence threshold for detecting text lines, ensuring that only high-confidence detections are considered.

• **Bounding Boxes:** The bounding boxes are sorted based on their y-coordinates to maintain the order of text lines. This sorting ensures that text lines are processed in their natural sequence, which is crucial for accurate text recognition and translation.

4. Prediction Function (Text Recognition)

The predict function encompasses several critical processes to recognize text from images. It integrates text line detection, cropping, and recognition into a seamless workflow.

4.1 Text Line Detection

- **Detection Results:** The YOLO model’s predict method generates bounding boxes around detected text lines. These boxes provide the regions of interest where text lines are located.
- **Sorting:** Bounding boxes are sorted to maintain the correct reading order of text lines. This sorting is essential for ensuring that the text is read and processed in the same sequence as it appears in the image.

4.2 Drawing Bounding Boxes

- **Visualization:** The ImageDraw module is used to draw bounding boxes around detected text lines, allowing users to visualize the detection results. Random colors are used for different boxes to enhance clarity and make it easier to distinguish between different text lines.

4.3 Cropping Detected Lines

- **Image Cropping:** Detected text lines are cropped from the original image. Each cropped image represents a single line of text and is stored in the cropped_images list. This step is crucial for isolating individual text lines for further processing and recognition.

4.4 Text Recognition

- OCR Processing: Each cropped image is processed by the text_recognizer function, which utilizes the OCR model to recognize the text. The recognized text is collected in the texts list. This function handles the core task of translating the cropped images into readable Urdu text.

4.5 Returning Results

- Output: The predict function returns both the input image with bounding boxes and the recognized text. This output allows users to see the detected text lines along with the corresponding recognized text, providing a comprehensive view of the OCR process.

5. Gradio Interface

A web interface is created using the Gradio library to facilitate user interaction with the OCR system. This interface enables users to upload images, view detection results, and see recognized text.

5.1 Gradio Interface Initialization

- Setup: The line iface = gr.Interface(predict, inputs=input, outputs=[output_image, output_text], title="Urdu Heritage Translation Project") initializes a Gradio interface. This interface provides an easy-to-use platform for users to interact with the OCR system. Users can upload images, view the processed results, and see the recognized text in a user-friendly format.

5.2 User Interaction

- **Functionality:** Users can upload images through the web interface, which triggers the predict function. The interface displays the processed image with bounding boxes and the recognized text, allowing users to easily test and apply the OCR system.

6. Text Recognition Function (text_recognizer)

The text_recognizer function handles the core text recognition process using the OCR model. It includes several steps to process images and extract text.

6.1 Image Processing

- **Grayscale Conversion:** The cropped image is converted to grayscale to simplify the recognition process. Grayscale conversion reduces the complexity of the image by removing color information, focusing on the text's shape and structure.
- **Resizing:** The image is resized to a standard width (e.g., 400 pixels) and height (32 pixels) to maintain uniformity. Resizing ensures that the input image dimensions are consistent, which is important for the OCR model's performance.

6.2 Normalization

- **Padding and Normalization:** The image is normalized using the NormalizePAD class to ensure consistent dimensions and padding. This normalization step prepares the image for input into the OCR model, ensuring that it meets the model's requirements.
- **Tensor Conversion:** The normalized image is transformed into a tensor format suitable for model input. This transformation is necessary for the OCR model to process the image and generate predictions.

6.3 Prediction

- Model Inference: The OCR model predicts the text from the processed image. The predicted indices are converted to actual text using the converter.decode() function. This decoding step translates the model's output into readable Urdu text, completing the recognition process.

7. Model Class

The Model class defines the architecture of the OCR model, including feature extraction, sequence modeling, and prediction components.

7.1 Feature Extraction

- **UNet Architecture:** The UNet_FeatureExtractor is used to extract features from the image. UNet is known for its effectiveness in image segmentation tasks, leveraging its encoder-decoder structure to capture detailed features from input images.

7.2 Sequence Modeling

- **Bidirectional LSTM:** The model employs Bidirectional LSTM layers to process the sequence of features. Bidirectional LSTM networks are beneficial for understanding the context of each character within the text sequence, as they process information from both forward and backward directions.

7.3 Prediction

- **Fully Connected Layer:** A fully connected layer is used to predict the final characters based on the processed sequence features. This layer maps the output of the sequence model to actual characters, providing the final text recognition output.

Conclusion

The OCR system described integrates a variety of advanced technologies and methodologies to achieve accurate recognition and translation of Urdu text from images. By leveraging a detailed character recognition system, a carefully configured model, and a robust YOLO detection model, the system efficiently processes and recognizes Urdu text.

The CTCLabelConverter ensures precise character mapping, while the YOLO detection model effectively identifies text lines in images. The predict function handles the entire workflow from

text detection to recognition, providing a seamless process from image input to text output. The Gradio interface enhances user interaction, allowing for easy testing and application of the OCR system.
The text_recognizer function, with its focus on image processing, normalization, and prediction, is central to the system's performance. The model architecture, including feature extraction with UNet and sequence modeling with Bidirectional LSTM, contributes significantly to the effectiveness of the text recognition process.
In summary, the OCR system's comprehensive approach combines state-of-the-art technology and advanced techniques to deliver a reliable and efficient solution for Urdu text recognition. The integration of these components ensures high accuracy and usability, making the system a valuable tool for text translation and document processing tasks.

Figure 10: Sample images from the Urdu Doc Dataset: An annotated real-world benchmark for Urdu text line detection in scanned document

3.4.1 FEATURE PREPARATION WORKFLOW

The feature preparation workflow for the Urdu Heritage Translation Project outlines the sequential steps involved in preparing historical Urdu documents for translation and modern use. The workflow consists of the following key stages:

1. Document Collection and Digitization

- Identify and gather historical Urdu documents from libraries, archives, and private collections.
- Scan the documents to convert them into high-resolution digital images.
- Organize the scanned documents into categorized datasets based on themes, periods, or sources.

2. Preprocessing

- Apply Optical Character Recognition (OCR) to extract text from scanned images.
- Clean and preprocess the extracted text by removing noise, correcting OCR errors, and formatting it into a structured digital format.
- Annotate the text with metadata, such as document origin, author, and historical context.

3. Feature Extraction

- Extract linguistic features such as word frequencies, sentence structures, and stylistic markers relevant for translation.
- Identify unique historical or cultural phrases that require specialized translation or annotation.
- Develop feature vectors for machine learning models if automated translation or text analysis tools are being employed.

4. Translation Preparation

- Segment the text into manageable units (e.g., sentences or paragraphs) for efficient translation.
- Identify and tag elements requiring special handling, such as idiomatic expressions or poetic structures.
- Prepare glossaries and reference materials to ensure consistency in translation.

5. Validation and Quality Assurance

- Verify the extracted and prepared features against the original documents to ensure accuracy.
- Conduct pilot translations to evaluate the effectiveness of the prepared features in guiding the translation process.
- Refine the workflow based on feedback and observed challenges.

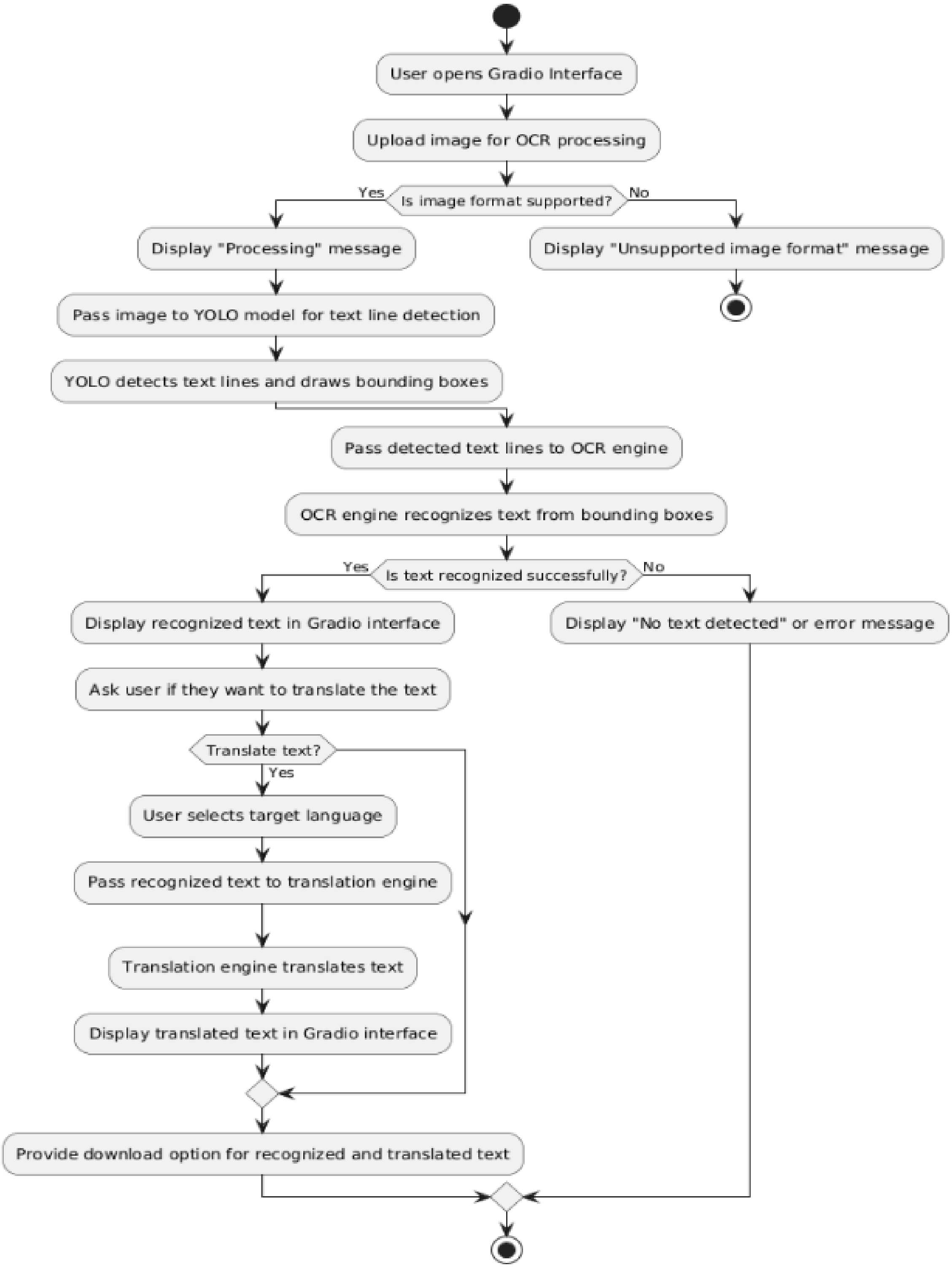

Figure 11: Overall Project Workflow

CHAPTER 4: RESULTS AND OUTPUTS

CHAPTER 4

RESULTS AND OUTPUTS

This section provides a detailed overview of the OCR system's performance, showcasing its interface, text line detection capabilities, and the output of recognized text. The following figures illustrate key aspects of the system and demonstrate its effectiveness.

Gradio Interface

The Gradio interface is an integral part of the OCR system, providing a user-friendly platform for interaction. This web-based interface allows users to upload images and view OCR results directly.

- **Description**: The Gradio interface includes an upload button for users to select images from their local devices. Once an image is uploaded, the system processes it and displays the results, including the recognized text and visual feedback.
- **Functionality**: The interface is designed to be intuitive, making it easy for users to test the OCR system without requiring any technical knowledge. It supports various image formats and provides real-time feedback on the processing status.

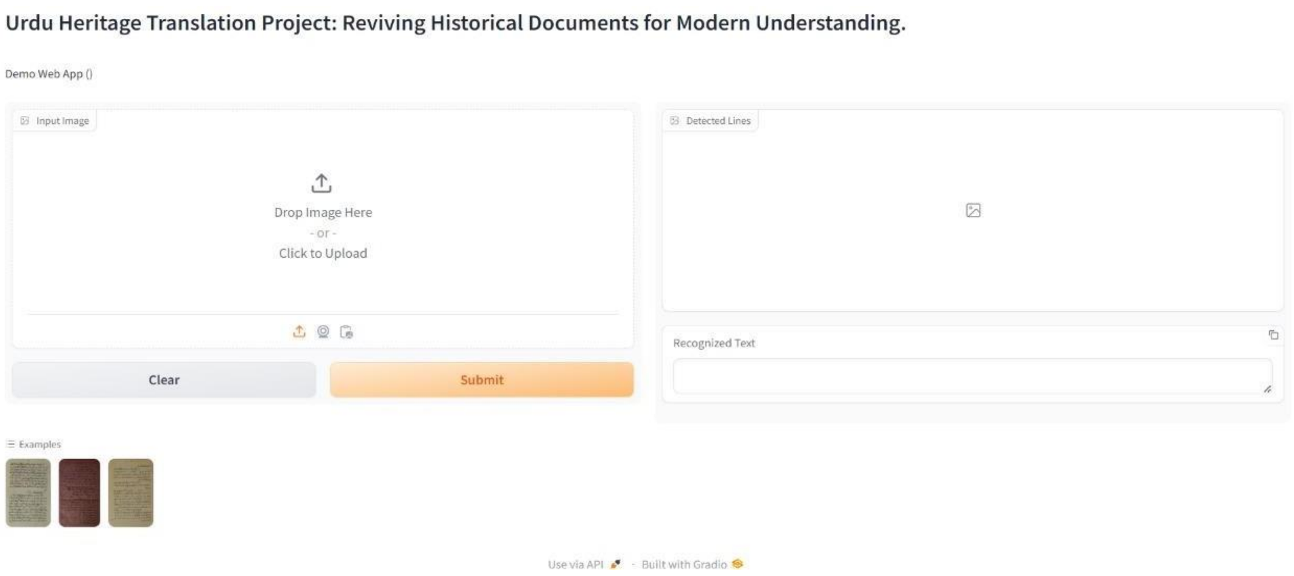

Fig 12: This screenshot shows the Gradio interface where users can upload images and view OCR results. The interface facilitates easy interaction and immediate display of processing results.

YOLO Model Bounding Boxes

The YOLO (You Only Look Once) model plays a crucial role in text line detection within the OCR system. This figure demonstrates how YOLO generates bounding boxes around detected text lines in the input image.

- **Description**: The image illustrates the bounding boxes created by the YOLO model, which identify and isolate text lines from the rest of the image. Each box highlights a specific region containing text, helping to segment the text lines for further processing.
- **Effectiveness**: The YOLO model's ability to detect text lines accurately is essential for the OCR system's performance. The bounding boxes ensure that text lines are properly segmented, reducing errors in text recognition.

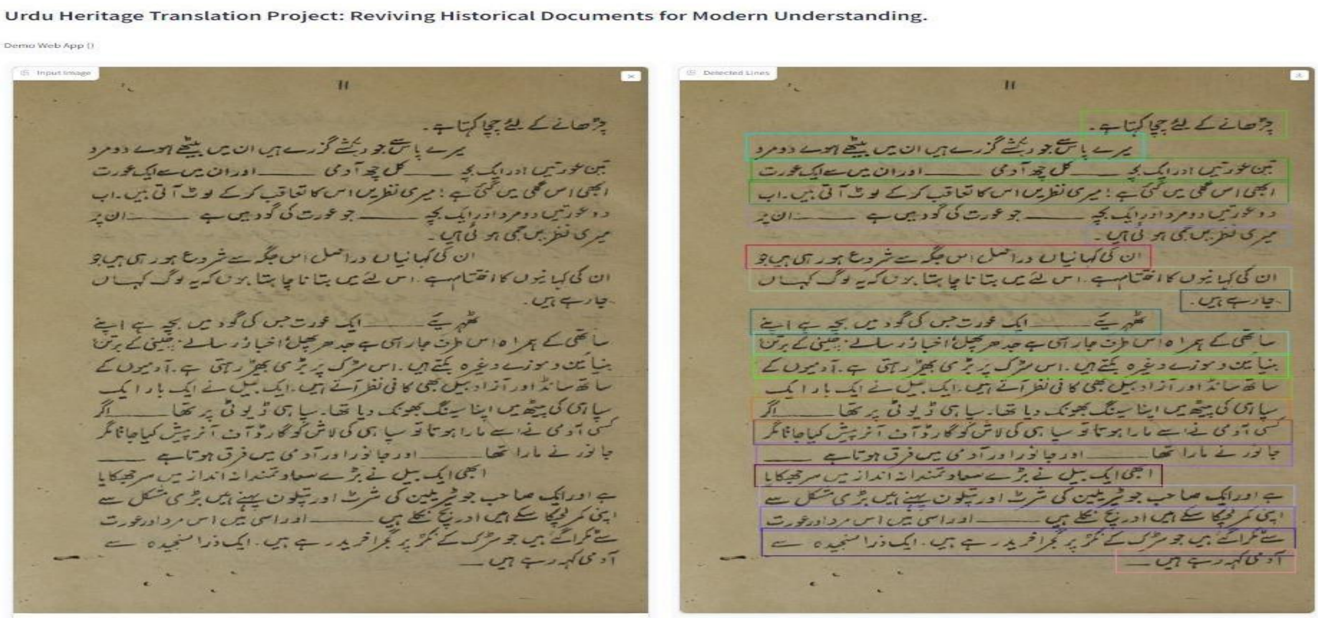

Fig 13: The image demonstrates the bounding boxes produced by the YOLO model around detected text lines. These boxes help in accurately isolating text regions for recognition.

Recognized Text Output

The final output image showcases the text that has been recognized and transcribed from the input image by the OCR system.

- **Description**: This figure displays the result of the OCR process, where the system has successfully identified and transcribed the Urdu text from the uploaded image. The recognized text is shown in a clear format, highlighting the OCR system's capability to convert visual text data into machine-readable text.
- **Performance**: The output reflects the effectiveness of the OCR system in handling Urdu text, demonstrating its ability to accurately process and extract information from document images.

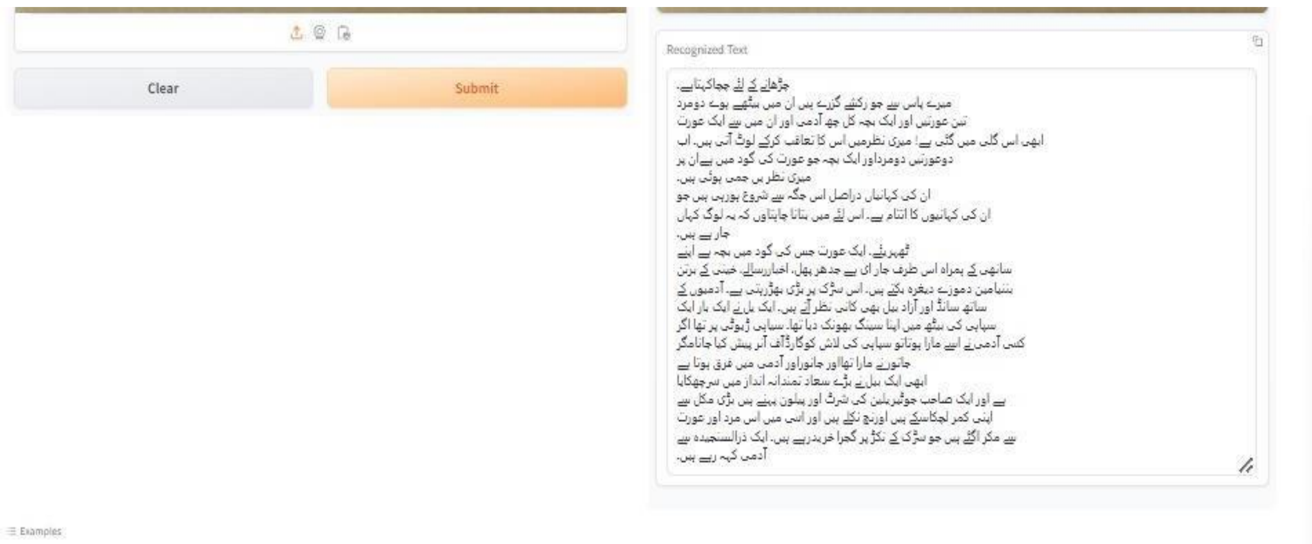

Fig 14: The output image displaying the recognized text extracted from the input image. capabilities.

CHAPTER 5: CONCLUSION

CHAPTER 5
CONCLUSION

5.1 Conclusion

The development and implementation of the Optical Character Recognition (OCR) system have reached significant milestones, with the core functionalities being successfully achieved. This section provides a summary of the project's progress and outlines the remaining tasks.
Key Achievements:
1.User Interface Development: The Gradio interface has been effectively designed and implemented, providing a user-friendly platform for interaction. Users can easily upload images and view the results of the OCR process.
2.Text Line Detection: The integration of the YOLO model has proven successful in detecting and segmenting text lines within images. This functionality is crucial for accurate text recognition, ensuring that each text line is properly isolated for processing.
3.Text Recognition: The OCR system has demonstrated its capability to accurately recognize and transcribe Urdu text from images. Preliminary tests show that the system can efficiently convert visual text data into machine-readable text.
Pending Work:
•Translation Module: The development of the translation module is still underway. This component is essential for providing full functionality, as it will enable the translation of recognized text into various target languages. Future tasks include implementing advanced machine translation models, integrating them with the existing OCR system, and conducting rigorous testing to ensure their accuracy and effectiveness.
In conclusion, the OCR system has made substantial progress, with successful implementation of key features and functionalities. The next phase involves completing the translation module to achieve the system’s full potential. Continued development and refinement will enhance the system's capabilities, providing a comprehensive solution for Urdu text recognition and translation.

5.2 Future Scope

The Urdu Heritage Translation Project has immense potential for future growth and impact. It can lead to the development of digital archives for Urdu heritage, making historical texts accessible globally. The techniques used in the project can be extended to preserve other regional languages. Advancements in AI and machine learning can further enhance OCR accuracy and translation quality. Additionally, the project can support the creation of educational tools and platforms, enabling broader engagement with and understanding of cultural and historical texts.

CHAPTER 6: REFERENCES

CHAPTER 6
REFERENCES

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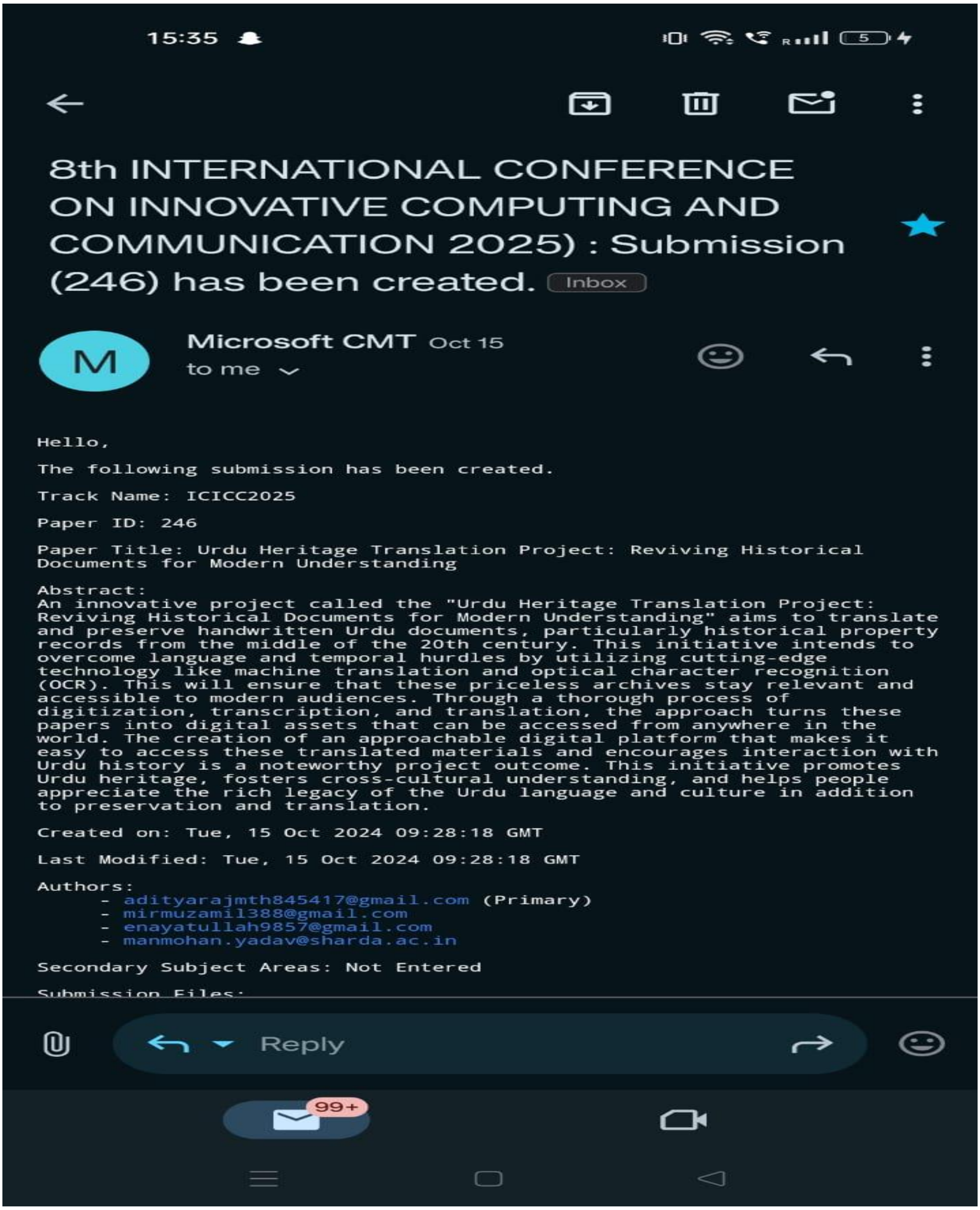
ANNEXURE-I



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Paper ID: 246

Paper Title: Urdu Heritage Translation Project: Reviving Historical Documents for Modern Understanding

Abstract:
An innovative project called the "Urdu Heritage Translation Project: Reviving Historical Documents for Modern Understanding" aims to translate and preserve handwritten Urdu documents, particularly historical property records from the middle of the 20th century. This initiative intends to overcome language and temporal hurdles by utilizing cutting-edge technology like machine translation and optical character recognition (OCR). This will ensure that these priceless archives stay relevant and accessible to modern audiences. Through a thorough process of digitization, transcription, and translation, the approach turns these papers into digital assets that can be accessed from anywhere in the world. The creation of an approachable digital platform that makes it easy to access these translated materials and encourages interaction with Urdu history is a noteworthy project outcome. This initiative promotes Urdu heritage, fosters cross-cultural understanding, and helps people appreciate the rich legacy of the Urdu language and culture in addition to preservation and translation.

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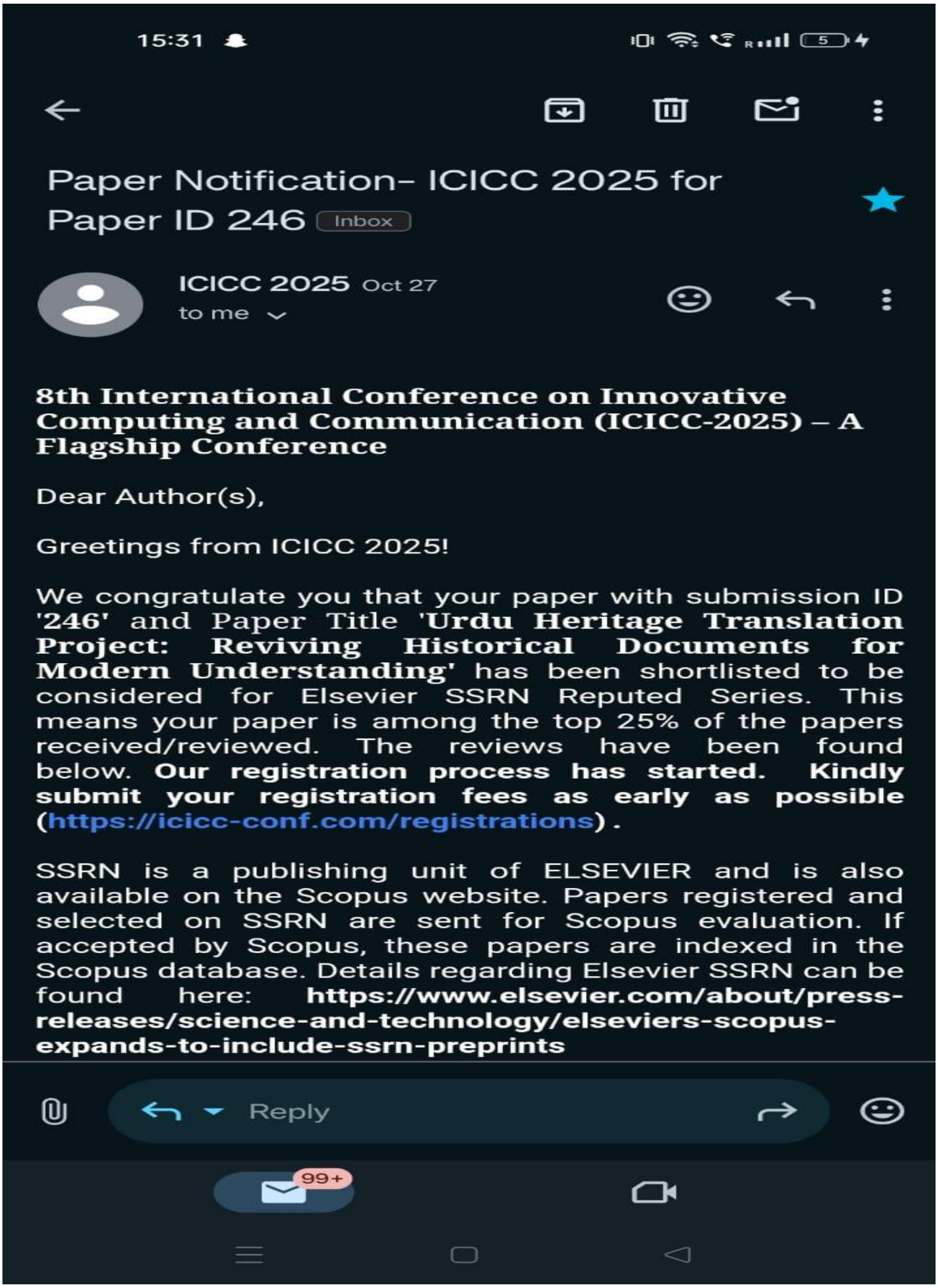
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